Structural Organisation in Animals: Organ Systems & Frog (*Rana tigrina*)

Exam Standard: NEET UG | Subject: Biology (Zoology)

Topic: Tissue to Organ System Level & Amphibian Morphology/Anatomy

1. Learning Objectives & Chapter Overview

In unicellular organisms, all vital life processes—such as digestion, respiration, and reproduction—are executed by a single cell. In contrast, multicellular animals possess a complex body where these fundamental functions are carried out by different groups of cells in a well-organised manner. This tutorial explores how tissues organize into organs and organ systems, and provides a comprehensive, exam-aligned study of the morphology and internal anatomy of the common Indian frog (*Rana tigrina*), representing vertebrates.

Key Competencies for NEET UG:

- Understand the biological hierarchy from cells to tissues, organs, and organ systems, and how this leads to a **division of labour**.
- Distinguish between **morphology** and **anatomy** in animals, plants, and microbes.
- Master the external diagnostic features, sexual dimorphism, and behavioural adaptations (camouflage, aestivation, hibernation) of *Rana tigrina*.
- Trace the complete structural and functional pathways of the digestive, respiratory, circulatory, lymphatic, excretory, neuro-endocrine, sensory, and reproductive systems in frogs.

2. Core Concepts: Tissue, Organ, and Organ System

2.1 Cellular Organisation and Division of Labour

Even simple multicellular organisms like *Hydra* are constructed of different types of cells, where the number of cells in each type can be in thousands. The human body is composed of billions of cells performing diverse functions. In multicellular animals, a group of similar cells along with intercellular substances performing a specific function is called a **tissue**.

Epithelia, for instance, are sheet-like tissues lining the body's surface and its cavities, ducts, and tubes. Epithelia possess one free surface facing either a body fluid or the outside environment, and their cells are structurally and functionally connected at junctions.

Remarkably, all complex animals consist of only **four basic types of tissues**:

1. **Epithelial tissue**
2. **Connective tissue**
3. **Muscular tissue**
4. **Neural tissue**

These four basic tissues are organised in specific proportions and patterns to form an **organ** such as the stomach, lung, heart, or kidney. Each organ in our body is made of one or more types of tissues. For example, **our heart consists of all four basic types of tissues**—epithelial, connective, muscular, and neural.

When two or more organs perform a common function through their physical and/or chemical interaction, they associate to form an **organ system** (e.g., digestive system, respiratory system). Cells, tissues, organs, and organ systems split up the physiological work in a way that exhibits a **division of labour** and ensures the survival of the body as a whole. Furthermore, the increasing complexity in organs and organ systems across different animal groups displays a discernible trend known as an **evolutionary trend**.

Cells ⟶ Tissues (Cells + Intercellular Substances) ⟶ Organs ⟶ Organ Systems ⟶ Whole Organism

2.2 Morphology vs. Anatomy

- **Morphology:** Refers to the study of form or externally visible features. In the case of plants or microbes, the term morphology precisely means only this. In the case of animals, morphology refers to the external appearance of the organs or parts of the body.
- **Anatomy:** Conventionally used for the study of the morphology of internal organs in animals.

3. Frogs (*Rana tigrina*): Habitat, Adaptations, and Morphology

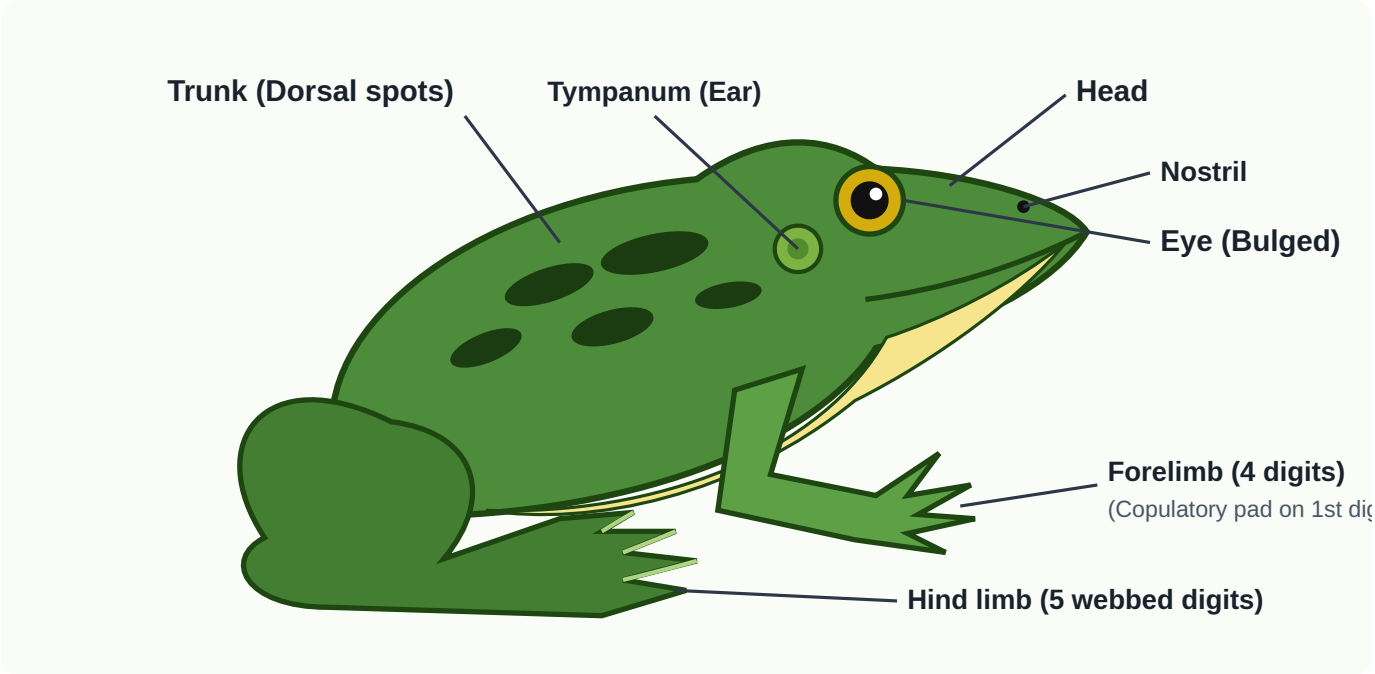
3.1 Taxonomic Position and Physiological Adaptations

Frogs are capable of living both on land and in freshwater. They belong to the **Class Amphibia** of **Phylum Chordata**. The most common species of frog (the Indian bullfrog) found in India is *Rana tigrina*.

- **Poikilothermy (Cold-bloodedness):** Frogs do not have a constant body temperature; their body temperature varies with the temperature of the environment. Such animals are called **cold-blooded** or **poikilotherms**.
- **Camouflage and Mimicry:** Frogs change their body colour while in grasses and on dry land to hide from their enemies (**camouflage**). This protective coloration is called **mimicry**.
- **Dormancy in Extreme Seasons:** Frogs are not seen during peak summer and winter. During these periods, they take shelter in deep burrows to protect themselves from extreme heat and cold. This is known as **summer sleep (aestivation)** and **winter sleep (hibernation)**, respectively.

3.2 External Features (Morphology)

- **Skin and Water Absorption:** The skin is smooth and slippery due to the presence of mucus secreted by mucous glands, and it is always maintained in a moist condition. It is also highly vascularised. The dorsal side of the body is generally **olive green with dark irregular spots**, while the ventral side is **uniformly pale yellow**. Crucially, **the frog never drinks water but absorbs it through the skin**.
- **Body Divisions:** The body of a frog is divisible into two main parts: **head** and **trunk**. A **neck and tail are absent**.
- **Structures on the Head:**
  - A pair of **nostrils** is present above the mouth.
  - **Eyes** are bulged and covered by a **nictitating membrane** that protects them while in water.
  - On either side of the eyes, a membranous **tympanum (ear)** receives sound signals. An external ear is absent.
- **Limbs and Locomotion:** The forelimbs and hind limbs help in **swimming, walking, leaping, and burrowing**. The **hind limbs end in five (5) digits** and are larger and more muscular than the **forelimbs, which end in four (4) digits**. The feet have **webbed digits** that assist in swimming.
- **Sexual Dimorphism:** Male and female frogs can be distinguished externally. Male frogs possess two diagnostic structures that are completely absent in female frogs:
  1. Sound-producing **vocal sacs**.
  2. A **copulatory pad on the first digit of the forelimbs**.



Tutorial Diagram 1: External features (morphology) of the Indian bullfrog (*Rana tigrina*). Note the absence of a neck and tail.

4. Internal Anatomy and Organ Systems of Frog

The body cavity of frogs accommodates well-developed organ systems: digestive, respiratory, circulatory, lymphatic, excretory, nervous, endocrine, sensory, and reproductive systems.

4.1 Digestive System

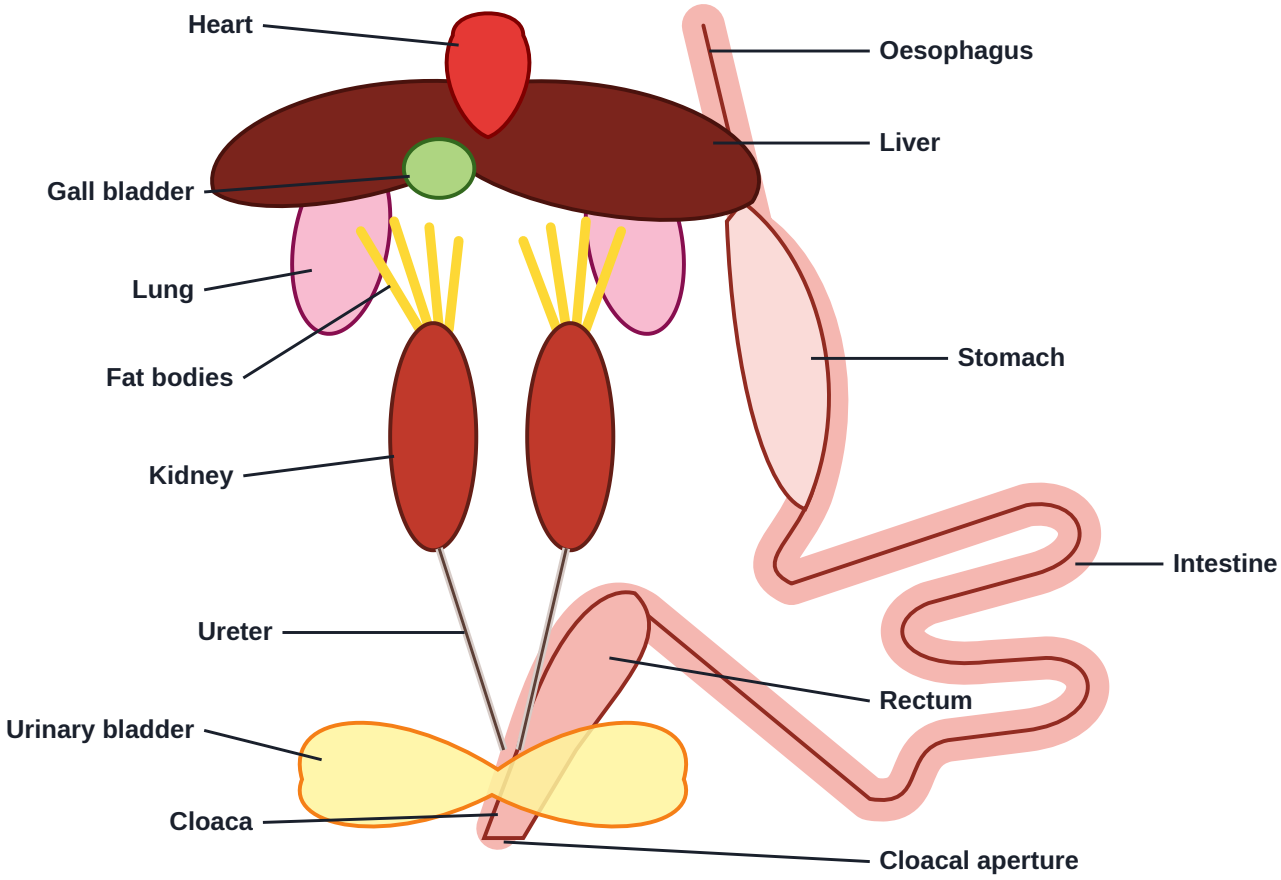
The digestive system comprises the **alimentary canal** and **digestive glands**. Because frogs are **carnivores**, the alimentary canal is short, and the length of the intestine is reduced.

- **Prey Capture:** Food is captured by a muscular **bilobed tongue** (bilobed at the tip).
- **Pathway of the Alimentary Canal:**

> **Mouth $\longrightarrow$ Buccal Cavity $\longrightarrow$ Pharynx $\longrightarrow$ Oesophagus (short tube) $\longrightarrow$ Stomach $\longrightarrow$ Intestine $\longrightarrow$ Rectum $\longrightarrow$ Cloaca $\longrightarrow$ Cloacal Aperture**

- **Digestive Glands and Ducts:**
  - **Liver:** Secretes **bile**, which is stored in the **gall bladder**.
  - **Pancreas:** A digestive gland that produces **pancreatic juice** containing digestive enzymes.
  - **Common Bile Duct:** The first part of the small intestine, the **duodenum**, receives both bile from the gall bladder and pancreatic juices from the pancreas through a **common bile duct**.

- **Physiology of Digestion and Absorption:**
  - In the **stomach**, digestion takes place by the action of HCl and gastric juices secreted from the walls of the stomach.
  - Partially digested acidic food, called **chyme**, passes from the stomach into the **duodenum**.
  - In the duodenum, **bile emulsifies fat**, and **pancreatic juices digest carbohydrates and proteins**.
  - **Final digestion** takes place in the intestine.
  - Digested food is absorbed by numerous finger-like folds in the inner wall of the intestine called **villi and microvilli**.
  - Undigested solid waste moves into the **rectum** and passes out to the exterior through the **cloaca**.



Tutorial Diagram 2: Diagrammatic representation of the internal viscera of a frog showing the complete digestive system and associated organs.

4.2 Respiratory System

Frogs respire both in water and on land using different methods:

Environment / Physiological State	Respiratory Organ(s) Involved	Mode & Mechanism of Respiration
In Water	Moist, vascularised **Skin**	**Cutaneous respiration:** Dissolved oxygen in water is exchanged directly through the skin by diffusion.
On Land (Active State)	**Buccal cavity**, **Skin**, and **Lungs**	**Pulmonary respiration** (via lungs), along with buccopharyngeal and cutaneous gaseous exchange.
During Aestivation & Hibernation	**Skin** only	**Cutaneous respiration:** Gaseous exchange takes place exclusively through the moist skin while sheltered in deep burrows.

Structure of Lungs & Air Pathway: The lungs are a pair of elongated, pink-coloured, sac-like structures situated in the upper part of the trunk region (thorax). Air enters through the **nostrils** $\longrightarrow$ **buccal cavity** $\longrightarrow$ **lungs**.

4.3 Vascular (Circulatory) and Lymphatic Systems

The vascular system of the frog is a well-developed **closed type** with **single circulation** (mixed blood is pumped from the single ventricle). Frogs also possess a **lymphatic system**.

A. Blood Vascular System (Heart, Blood Vessels, and Blood)

- **Heart Structure:** A muscular structure situated in the upper part of the body cavity, enclosed by a protective membrane called the **pericardium**.
- **Chambers of the Heart:** It is **three-chambered**, consisting of **two atria (right and left)** and **one ventricle**.
- **Accessory Chambers:**

- **Sinus venosus:** A **triangular structure** that joins the **right atrium** (dorsally). It receives deoxygenated blood through the major veins called **vena cava**.
- **Conus arteriosus:** The ventricle opens into a **sac-like structure** called the conus arteriosus on the **ventral side** of the heart.

- **Arterial and Venous Systems:** Blood from the heart is distributed to all body parts by arteries (**arterial system**), while veins collect blood from body parts back to the heart (**venous system**).
- **Special Venous Connections (Portal Systems):** Frogs possess **both** portal systems:
  1. **Hepatic portal system:** Special venous connection between the **liver and intestine**.
  2. **Renal portal system:** Special venous connection between the **kidney and lower parts of the body**.

- **Composition of Blood:** Composed of liquid **plasma** and blood cells—**erythrocytes (RBCs)**, **leucocytes (WBCs)**, and **platelets**.
  - **Important NEET Fact:** Frog **RBCs are nucleated** and contain the red-coloured respiratory pigment **haemoglobin** (unlike mature mammalian RBCs, which are enucleated).

B. Lymphatic System

- Consists of **lymph, lymph channels, and lymph nodes**.
- **Difference between Lymph and Blood:** Lymph is different from blood because **it lacks a few proteins and RBCs**.

4.4 Excretory System

The elimination of nitrogenous wastes is performed by a well-developed excretory system consisting of a **pair of kidneys, ureters, cloaca, and urinary bladder**.

- **Kidneys:** Compact, dark red, and bean-like structures situated a little posteriorly in the body cavity on both sides of the vertebral column. Each kidney is composed of several structural and functional units called **uriniferous tubules** or **nephrons**.
- **Ureters (Sex-Specific Function):**
  - In **male frogs**, two ureters emerge from the kidneys and act as a **urinogenital duct** (carrying both urine and spermatozoa), opening into the cloaca.
  - In **female frogs**, the **ureters** (carrying only urine) and the **oviducts** (carrying ova) **open separately into the cloaca**.

- **Urinary Bladder:** A thin-walled structure present **ventral to the rectum**, which also opens into the cloaca.
- **Nature of Nitrogenous Waste:** The adult frog excretes **urea** and is therefore a **ureotelic animal**. Excretory wastes are carried by blood into the kidney, where they are separated and excreted.

4.5 System for Control and Coordination (Neural & Endocrine)

Control and coordination are highly evolved in the frog, comprising both the **neural system** and **endocrine glands**.

A. Endocrine System

Chemical coordination of various organs is achieved by hormones secreted by endocrine glands. The prominent endocrine glands found in frogs are:

- Pituitary, Thyroid, Parathyroid, Thymus, Pineal body, Pancreatic islets, Adrenals, and Gonads.

B. Nervous System

The nervous system is organised into three divisions:

1. **Central Nervous System (CNS):** Brain and spinal cord.
2. **Peripheral Nervous System (PNS):** Cranial and spinal nerves. There are **ten (10) pairs of cranial nerves** arising from the brain.
3. **Autonomic Nervous System (ANS):** Sympathetic and parasympathetic systems.

Structure of the Brain: Enclosed in a bony structure called the **brain box (cranium)**, the brain is divided into three major regions:

Brain Division	Component Structures
Fore-brain	Olfactory lobes, paired cerebral hemispheres, and unpaired diencephalon.
Mid-brain	Characterised by a pair of **optic lobes**.
Hind-brain	**Cerebellum** and **medulla oblongata**.

The **medulla oblongata** passes out through the **foramen magnum** and continues into the **spinal cord**, which is enclosed in the vertebral column.

4.6 Sense Organs

Frogs possess five different types of sense organs. Out of these, only the **eyes and internal ears are well-organised structures**, whereas the remaining three are merely **cellular aggregations around nerve endings**:

Sensory Modality	Sense Organ in Frog	Level of Structural Organisation
Touch	Sensory papillae	Cellular aggregations around nerve endings
Taste	Taste buds	Cellular aggregations around nerve endings
Smell	Nasal epithelium	Cellular aggregations around nerve endings
Vision	Eyes (pair of spherical structures in skull orbit; **simple eyes** possessing only one unit)	**Well-organised structures**
Hearing & Equilibrium	Tympanum with internal ears (external ear is absent; only tympanum is visible externally)	**Well-organised structures** (functions in both **hearing and balancing/equilibrium**)

4.7 Reproductive Systems and Development

Frogs have well-organised male and female reproductive systems.

A. Male Reproductive System

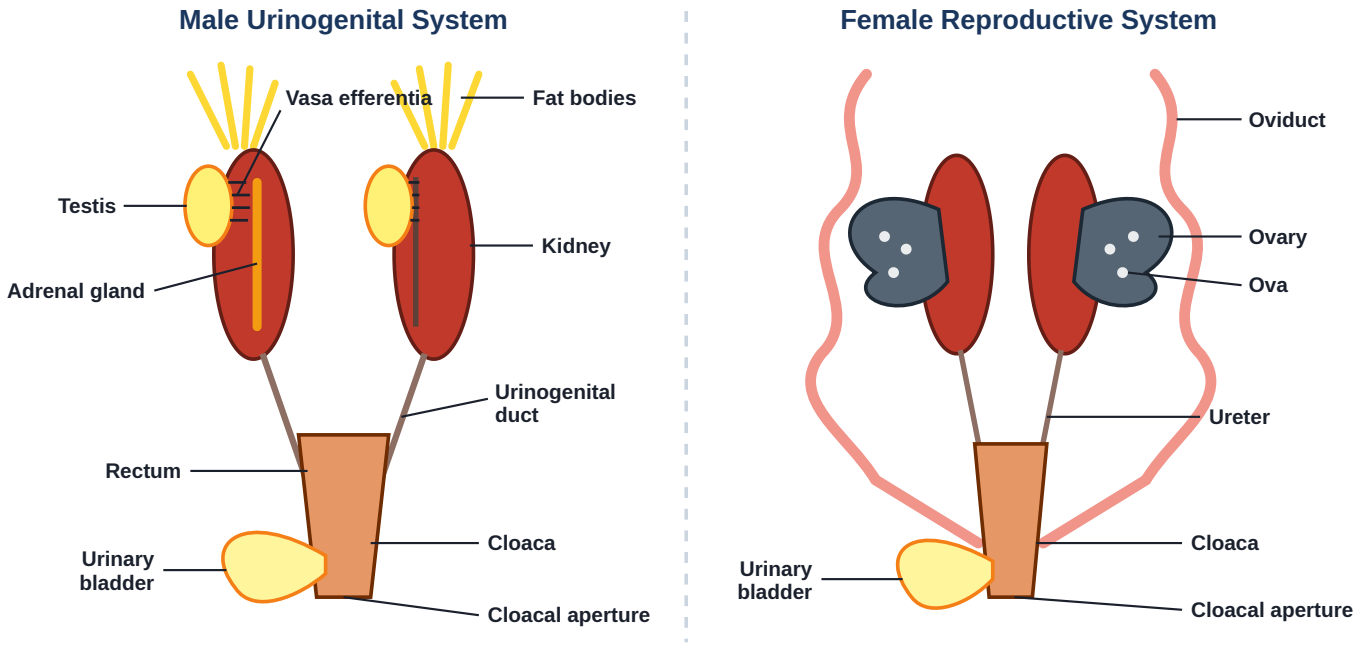
- **Testes:** A pair of yellowish, ovoid testes found adhered to the upper part of the kidneys by a double fold of peritoneum called **mesorchium**.
- **Vasa efferentia: 10–12 in number**; they arise from the testes, enter the kidneys on their side, and open into **Bidder's canal**.
- **Pathway of Spermatozoa:**

  Testes ⟶ Vasa efferentia (10–12) ⟶ Bidder's canal (inside kidney) ⟶ Urinogenital duct (Ureter) ⟶ Cloaca ⟶ Cloacal Aperture

- **Cloaca:** A small, median chamber used to pass **faecal matter, urine, and sperms** to the exterior.

B. Female Reproductive System

- **Ovaries:** A pair of ovaries situated near the kidneys. Crucially, **there is no functional connection between the ovaries and the kidneys**.
- **Oviducts:** A pair of oviducts arising from the ovaries opens into the **cloaca separately** (distinct from the ureters).
- **Oviposition, Fertilisation, and Development:**
  - A mature female can lay **2500 to 3000 ova at a time**.
  - **Fertilisation is external** and takes place in **water**.
  - Development is **indirect** and involves an aquatic larval stage called a **tadpole**. The tadpole undergoes **metamorphosis** to form the adult frog.



Tutorial Diagram 3: Comparison of the Male Reproductive System (left) and Female Reproductive System (right) of a frog. Note that vasa efferentia connect the testes to the kidneys in males, whereas ovaries have no functional connection with kidneys in females.

4.8 Ecological and Economic Importance of Frogs

- **Crop Protection:** Frogs are beneficial for mankind because they eat insects and protect crops from pests.
- **Ecological Balance:** Frogs maintain ecological balance because they serve as an important link in the food chain and food web in the ecosystem.
- **Human Food Source:** In some countries, the muscular legs of frogs are used as food by humans.

5. High-Yield Comparative Summary Tables for NEET

5.1 Male vs. Female Frog (*Rana tigrina*)

Feature	Male Frog	Female Frog
Vocal Sacs	Present (sound-producing).	Absent.
Copulatory Pad	Present on the **first digit of the forelimbs**.	Absent.
Gonads	Pair of yellowish ovoid **testes**.	Pair of lobulated **ovaries**.
Connection of Gonads with Kidney	**Functional connection present:** Testes are adhered to upper kidneys by **mesorchium**; 10–12 **vasa efferentia** enter the kidney and open into **Bidder's canal**.	**No functional connection** between ovaries and kidneys.
Role of Ureters	Act as **urinogenital ducts** (transport both urine and sperms to cloaca).	Act strictly as **urinary ducts** (transport only urine; oviducts open separately into cloaca).

5.2 Quantitative Facts & Numbers to Memorise

Anatomical / Physiological Parameter	Exact Value / Count in Frog
Number of basic animal tissue types (all present in the heart)	4 (Epithelial, Connective, Muscular, Neural)
Digits in each forelimb	4 digits
Digits in each hind limb	5 webbed digits
Chambers in the frog heart	3 chambers (2 atria + 1 ventricle)
Cranial nerves arising from the brain	10 pairs
Vasa efferentia arising from each testis	$10–12$ in number
Ova laid by a mature female frog at a time	2500 to 3000 ova

6. Worked Examples & Conceptual Discussions

Worked Example 1: Structural & Functional Role of Ureters in Frogs

Problem Statement: Mention and compare the exact physiological function of the ureters in male and female frogs.

Solution:

- In **male frogs**, two ureters emerge from the kidneys and function as **urinogenital ducts**. Because the vasa efferentia from the testes enter the kidneys and open into Bidder's canal (which communicates with the ureter), the male ureter transports **both urine and spermatozoa** into the cloaca.
- In **female frogs**, the ovaries have no functional connection with the kidneys. Therefore, the female ureters function solely to conduct **urine** from the kidneys to the cloaca, while the oviducts open separately into the cloaca to release ova.

Worked Example 2: Tracing Anatomical Pathways

Problem Statement: A red blood cell is present in the capillaries of the frog's intestine, and a spermatozoon is produced in the frog's testis. Trace (a) the special venous pathway taken by the blood from the intestine before returning to the heart, and (b) the exact sequence of ducts traversed by the spermatozoon before exiting the body.

Solution:

1. **Venous Pathway from Intestine:** Frogs possess a special venous connection between the intestine and the liver called the **hepatic portal system**. Blood from the intestine first passes to the liver via the hepatic portal vein, and then from the liver through the systemic venous circulation (vena cava) into the triangular **sinus venosus**, which empties into the **right atrium** of the heart.
2. **Pathway of Spermatozoon:** Testis $\longrightarrow$ Vasa efferentia ($10–12$ in number) $\longrightarrow$ Bidder's canal (inside kidney) $\longrightarrow$ Urinogenital duct (ureter) $\longrightarrow$ Cloaca $\longrightarrow$ Cloacal aperture (to exterior).

7. NEET UG Practice Questions with Worked Solutions

Note: The following original practice questions are designed strictly in accordance with the NEET UG examination pattern (Single-Correct MCQs, Multi-Statement Analysis, Assertion–Reason, and Matrix-Match formats) to test conceptual clarity and factual precision.

Practice Question 1 (Single-Correct MCQ)

Which of the following statements correctly describes the composition of tissues in the human heart and the definition of animal anatomy?

(A) The heart is composed exclusively of muscular and connective tissues; anatomy is the study of externally visible features in microbes.

(B) The heart consists of all four basic types of tissues (epithelial, connective, muscular, and neural); anatomy conventionally refers to the study of the morphology of internal organs in animals.

(C) The heart consists of muscular, connective, and neural tissues but lacks epithelial tissue; morphology refers to the study of internal viscera.

(D) Only organs of the digestive system contain all four basic tissue types; anatomy is synonymous with external morphology in plants.

Correct Answer: (B)

Explanation: Each organ in our body is made of one or more types of tissues. Specifically, our heart consists of all four basic types of tissues: epithelial, connective, muscular, and neural. Furthermore, while morphology refers to externally visible features, the word anatomy is conventionally used for the study of the morphology of internal organs in animals.

Practice Question 2 (Single-Correct MCQ)

During peak summer and winter, *Rana tigrina* takes shelter in deep burrows to protect itself from extreme heat and cold. How does gaseous exchange occur during these periods of aestivation and hibernation?

(A) Exclusively through pulmonary respiration via elongated pink lungs

(B) Through the buccal cavity lining and nostrils by active ventilation

(C) Exclusively through the moist, vascularised skin (cutaneous respiration)

(D) No gaseous exchange occurs until the frog emerges from the burrow

Correct Answer: (C)

Explanation: During summer sleep (aestivation) and winter sleep (hibernation), frogs remain inactive in deep burrows and do not use lungs or buccal respiration. Gaseous exchange during aestivation and hibernation takes place solely through the moist skin (cutaneous respiration).

Practice Question 3 (Multi-Statement Evaluation)

Read the following statements regarding the external morphology of the Indian bullfrog (*Rana tigrina*):

1. The body is divisible into head, neck, and trunk, while a tail is present only in the adult stage.
2. The frog never drinks water but absorbs it through its moist skin.
3. The forelimbs end in five webbed digits, whereas the hind limbs end in four muscular digits.
4. Male frogs are distinguished from female frogs by the presence of vocal sacs and a copulatory pad on the first digit of the forelimbs.
5. Eyes are bulged and covered by a nictitating membrane that protects them while in water.

Which of the above statements are **correct**?

(A) 1, 2, and 4 only

(B) 2, 4, and 5 only

(C) 2, 3, 4, and 5 only

(D) 1, 3, and 5 only

Correct Answer: (B)

Explanation: Statements 2, 4, and 5 are correct. Statement 1 is incorrect because both a neck and a tail are absent in an adult frog (the body is divisible only into head and trunk). Statement 3 is incorrect because the hind limbs are larger and muscular and end in five (5) digits, whereas the forelimbs end in four (4) digits.

Practice Question 4 (Single-Correct MCQ)

Why is the alimentary canal of a frog relatively short with a reduced length of the intestine?

(A) Because frogs are herbivores and cellulose digestion occurs in the cloaca

(B) Because frogs are carnivores

(C) Because protein digestion is completed entirely in the buccal cavity

(D) Because frogs lack a duodenum and rectum

Correct Answer: (B)

Explanation: The alimentary canal of a frog is short because frogs are carnivores (feeding primarily on insects and small prey), which requires a shorter intestinal length for digestion compared to herbivorous diets.

Practice Question 5 (Single-Correct MCQ)

In the digestive system of *Rana tigrina*, partially digested food called chyme passes from the stomach into which structure, and what secretions does it receive there?

(A) Rectum; receives HCl and gastric juice via the cystic duct

(B) Duodenum; receives bile from the gall bladder and pancreatic juices from the pancreas through a common bile duct

(C) Oesophagus; receives bile directly from the liver and mucus from the pharynx

(D) Cloaca; receives pancreatic juice and intestinal villi secretions separately

Correct Answer: (B)

Explanation: Partially digested food called chyme passes from the stomach to the first part of the small intestine, the duodenum. The duodenum receives bile from the gall bladder and pancreatic juices from the pancreas through a common bile duct. Bile emulsifies fats, and pancreatic juices digest carbohydrates and proteins.

Practice Question 6 (Single-Correct MCQ)

Which of the following accurately describes the accessory chambers and portal systems of the frog's circulatory system?

(A) Sinus venosus joins the left atrium; conus arteriosus arises from the right atrium; renal portal system is absent.

(B) A triangular sinus venosus receives blood via vena cava and joins the right atrium; the ventricle opens into a sac-like conus arteriosus on the ventral side; both hepatic and renal portal systems are present.

(C) Conus arteriosus receives blood from vena cava; sinus venosus arises from the ventricle; only hepatic portal system is present.

(D) The heart has four chambers with complete double circulation; RBCs are enucleated and lack haemoglobin.

Correct Answer: (B)

Explanation: In the 3-chambered frog heart (two atria, one ventricle), a triangular structure called the sinus venosus joins the right atrium and receives blood through major veins called vena cava. The ventricle opens into a sac-like conus arteriosus on the ventral side of the heart. Furthermore, both hepatic portal system (between liver and intestine) and renal portal system (between kidney and lower parts of the body) are present.

Practice Question 7 (Single-Correct MCQ)

How does the lymph of a frog differ in composition from its blood?

(A) Lymph contains nucleated RBCs and haemoglobin, whereas blood lacks RBCs.

(B) Lymph lacks a few proteins and red blood cells (RBCs).

(C) Lymph contains platelets and erythrocytes but completely lacks water and leucocytes.

(D) Lymph and blood have identical cellular and protein compositions in amphibians.

Correct Answer: (B)

Explanation: Blood in frogs is composed of plasma and blood cells (nucleated RBCs containing haemoglobin, WBCs, and platelets). The lymph is different from blood because it lacks a few proteins and RBCs.

Practice Question 8 (Assertion–Reason)

Assertion (A): In male frogs, the ureters are referred to as urinogenital ducts, whereas in female frogs they are not.

Reason (R): In male frogs, vasa efferentia from the testes enter the kidneys and open into Bidder's canal which communicates with the ureter, whereas in females the ovaries have no functional connection with the kidneys and the oviducts open separately into the cloaca.

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) (A) is false, but (R) is true.

Correct Answer: (A)

Explanation: Both Assertion and Reason are completely accurate, and Reason directly explains Assertion. In males, 10–12 vasa efferentia arise from the testes, enter the kidneys on their side, and open into Bidder's canal, which communicates with the urinogenital duct (ureter) carrying both urine and sperm to the cloaca. In females, the ovaries have no functional connection with the kidneys, and the ureters and oviducts open separately into the cloaca.

Practice Question 9 (Matrix Match)

Match the divisions of the frog's nervous system in **Column I** with their respective structures or features in **Column II**:

Column I (Division)	Column II (Structures / Features)
(p) Fore-brain	(i) Cerebellum and medulla oblongata
(q) Mid-brain	(ii) 10 pairs arising from the brain
(r) Hind-brain	(iii) Olfactory lobes, paired cerebral hemispheres, and unpaired diencephalon
(s) Cranial nerves	(iv) A pair of optic lobes

(A) (p)–(iii), (q)–(iv), (r)–(i), (s)–(ii)

(B) (p)–(iv), (q)–(iii), (r)–(ii), (s)–(i)

(C) (p)–(iii), (q)–(i), (r)–(iv), (s)–(ii)

(D) (p)–(ii), (q)–(iv), (r)–(i), (s)–(iii)

Correct Answer: (A)

Explanation:
- Fore-brain includes olfactory lobes, paired cerebral hemispheres, and unpaired diencephalon: (p)–(iii).
- Mid-brain is characterised by a pair of optic lobes: (q)–(iv).
- Hind-brain consists of cerebellum and medulla oblongata: (r)–(i).
- Cranial nerves are 10 pairs arising from the brain: (s)–(ii).

Practice Question 10 (Single-Correct MCQ)

Out of the five types of sense organs found in *Rana tigrina*, which ones are **well-organised structures** rather than simple cellular aggregations around nerve endings?

(A) Sensory papillae and taste buds

(B) Nasal epithelium and sensory papillae

(C) Eyes and internal ears

(D) Taste buds, nasal epithelium, and external pinnae

Correct Answer: (C)

Explanation: Frogs have organs of touch (sensory papillae), taste (taste buds), smell (nasal epithelium), vision (eyes), and hearing (tympanum with internal ears). Out of these, only the **eyes and internal ears** are well-organised structures, while the rest (sensory papillae, taste buds, and nasal epithelium) are cellular aggregations around nerve endings.

Practice Question 11 (Single-Correct MCQ)

Identify the correct sequence of structures through which spermatozoa pass in a male frog from the site of production to the exterior:

(A) Testis $\longrightarrow$ Mesorchium $\longrightarrow$ Bidder's canal $\longrightarrow$ Vasa efferentia $\longrightarrow$ Cloaca

(B) Testis $\longrightarrow$ Vasa efferentia $\longrightarrow$ Bidder's canal $\longrightarrow$ Urinogenital duct $\longrightarrow$ Cloaca $\longrightarrow$ Cloacal aperture

(C) Testis $\longrightarrow$ Bidder's canal $\longrightarrow$ Vasa efferentia $\longrightarrow$ Urinary bladder $\longrightarrow$ Cloaca

(D) Testis $\longrightarrow$ Vasa efferentia $\longrightarrow$ Oviduct $\longrightarrow$ Rectum $\longrightarrow$ Cloacal aperture

Correct Answer: (B)

Explanation: Vasa efferentia (10–12 in number) arise from the testes, enter the kidneys on their side, and open into Bidder's canal. Bidder's canal communicates with the urinogenital duct (ureter) that emerges from the kidney and opens into the cloaca, which expels sperms through the cloacal aperture. (Note: Mesorchium is the double fold of peritoneum attaching the testis to the kidney, not a lumen for sperm transport).

Practice Question 12 (Multi-Statement Evaluation)

Consider the following statements regarding reproduction, excretion, and development in *Rana tigrina*:

1. An adult frog excretes urea and is therefore a ureotelic animal.
2. The thin-walled urinary bladder is located dorsal to the rectum and opens directly to the exterior independent of the cloaca.
3. A mature female frog can lay $2500 \text{ to } 3000$ ova at a time.
4. Fertilisation is internal and takes place inside the cloaca before the eggs are laid in water.
5. The ear of a frog serves as an organ of both hearing and balancing (equilibrium).

Which of the above statements are **correct**?

(A) 1, 3, and 5 only

(B) 1, 2, and 3 only

(C) 2, 4, and 5 only

(D) 1, 3, 4, and 5 only

Correct Answer: (A)

Explanation: Statements 1, 3, and 5 are correct. Statement 2 is incorrect because the thin-walled urinary bladder is present **ventral** to the rectum and opens into the cloaca. Statement 4 is incorrect because fertilisation in frogs is **external** and takes place in water.

8. Common NEET Pitfalls & Revision Checklist

Common Exam Pitfalls to Avoid

- **Pitfall 1 — Digit Count Reversal:** Students frequently swap the digit counts of the limbs. Remember: **Forelimbs = 4 digits** (with copulatory pad on the 1st digit in males); **Hind limbs = 5 webbed digits** (larger and more muscular).
- **Pitfall 2 — Mammalian vs. Amphibian RBCs:** Unlike human RBCs, **frog RBCs are nucleated** and contain haemoglobin. Also remember that frog lymph lacks a few proteins and RBCs.
- **Pitfall 3 — Sinus Venosus vs. Conus Arteriosus:** Do not confuse their locations! **Sinus venosus** (triangular, dorsal) receives vena cava and joins the **right atrium**. **Conus arteriosus** (sac-like, ventral) receives blood pumped out of the **ventricle**.
- **Pitfall 4 — Cranial Nerve Number:** Frogs (amphibians, anamniotes) have **10 pairs of cranial nerves**—do not confuse this with the 12 pairs found in reptiles, birds, and mammals.
- **Pitfall 5 — Mesorchium vs. Vasa Efferentia: Mesorchium** is the double fold of peritoneum that *adheres* the testis to the kidney; **vasa efferentia** (10–12) are the actual *ducts* that carry sperm from the testis into Bidder's canal inside the kidney.
- **Pitfall 6 — Ovary-Kidney Connection:** While testes have a direct structural and functional ductal connection with the kidneys in males, **ovaries have NO functional connection with the kidneys** in females.

Pre-Exam Rapid Revision Checklist

- ☑ All 4 basic tissue types (epithelial, connective, muscular, neural) are present in the **heart**.
- ☑ *Rana tigrina*: Class Amphibia, Phylum Chordata; poikilothermic; shows camouflage (mimicry), aestivation (summer sleep), and hibernation (winter sleep).
- ☑ Never drinks water; absorbs water through moist, mucous-covered skin.
- ☑ Alimentary canal is short (carnivorous); bilobed tongue captures prey; common bile duct brings bile + pancreatic juice to duodenum.
- ☑ Respiration: Cutaneous in water & during aestivation/hibernation; cutaneous, buccopharyngeal, and pulmonary on land.
- ☑ Closed vascular system with single circulation; 3-chambered heart; both hepatic and renal portal systems present.
- ☑ Ureotelic adult; kidney has nephrons (uriniferous tubules); urinary bladder is ventral to rectum.
- ☑ Brain: Forebrain (olfactory lobes, paired cerebral hemispheres, unpaired diencephalon), Midbrain (paired optic lobes), Hindbrain (cerebellum, medulla oblongata passing through foramen magnum).
- ☑ Well-organised sense organs: Eyes (simple) and Internal ears (hearing + equilibrium). Cellular aggregations: Sensory papillae, taste buds, nasal epithelium.
- ☑ External fertilisation in water; $2500–3000$ ova laid at a time; aquatic tadpole larva undergoes metamorphosis into adult.

9. Concise Final Recap

In multicellular animals, similar cells and intercellular substances form tissues, which organise in specific proportions to form organs, and interacting organs form organ systems—achieving a division of labour essential for survival. The Indian bullfrog (*Rana tigrina*) exemplifies vertebrate structural organisation. Adapted to both terrestrial and freshwater life, its moist vascular skin enables water absorption and cutaneous respiration, while its internal cavity houses well-coordinated digestive, pulmonary, closed single-circuit

vascular, portal, lymphatic, ureotelic excretory, neuro-endocrine, sensory, and reproductive systems. Together, these systems illustrate how cellular and tissue-level specialization supports complex physiological life and ecological balance.